

Portfolio Optimization

MSCF Investments

Fall 2021 Mini 1

- ❑ Portfolio math
- ❑ Factor structure of portfolio returns
- ❑ Decompositions

- ❑ Constant Proportion Portfolio Insurance (CPPI)
- ❑ Empirical evidence: Does trading create good returns?

Learning Objectives

After this lecture, you should be able to ...

- ❑ Construct **best probability distributions** for portfolio returns via security holdings decisions
- ❑ Understand investor **preferences over affordable portfolios**
- ❑ Decompose portfolio returns using **factor structure** of returns
- ❑ Decompose investment performance into parts due to **strategic, dynamic, & security-selection**
- ❑ Execute specific portfolio strategies

Probability and Statistics

Expectations of random variables

- Expectation of a sum: $E[x+y] = E[x] + E[y]$
- Expectation of a product with a constant: $E[ax] = a E[x]$
- Expectation of a product of two random variables: $E[xy] = E[x] E[y] + \text{cov}(x,y)$
- Iterated expectations: $E_t[E_s[x]] = E_t[x]$ where $s > t$
- Sample average $\sum_{i=1,\dots,N} x_i/N$ = Best linear unbiased (BLUE) estimate of expectation $E[x]$

Variances

- Definition: $\text{var}[x] = E[(x - E[x])^2]$
- Standard deviation: $\text{SD}[x] = \sqrt{\text{var}(x)}$
- Product with a constant: $\text{var}[ax] = a^2 \text{var}[x] \rightarrow \text{SD}[ax] = |a| \text{SD}[x]$
- Variance of a sum: $\text{var}[x+y] = \text{var}[x] + \text{var}[y] + 2\text{cov}(x,y)$

$$\rightarrow \text{var}[x_1 + \dots + x_N] = \sum_{i=1,\dots,N} \text{var}[x_i] + \sum_{j=1,\dots,N} \sum_{i \neq j} \text{cov}(x_j, x_i)$$

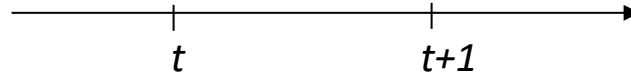
- Sample estimate: $\sum_{i=1,\dots,N} (x_i - \text{sample average})^2 / (N-1)$

Covariances

- Definition: $\text{cov}(x, y) = E[(x - E[x])(y - E[y])]$
- Covariance of a sum: $\text{cov}(x, y + z) = \text{cov}(x, y) + \text{cov}(x, z)$
- Covariance of products: $\text{cov}(ax, by) = ab \text{cov}(x, y)$
- Correlation: $\text{cov}(x, y) = \rho(x, y) \text{SD}[x] \text{SD}[y] \rightarrow \rho(x, y) = \text{cov}(x, y) / \text{SD}[x] \text{SD}[y]$

Definitions and Timing

Timing:



□ $P_{j,t}$ = Market price at t of the future random variable $P_{j,t+1} + CF_{j,t+1}$ at $t+1$

□ Return between t and $t+1$: $r_{jt+1} = \frac{P_{jt+1} + CF_{jt+1} - P_{jt}}{P_{jt}} = \frac{P_{jt+1} + CF_{jt+1}}{P_{jt}} - 1$

Question: Why focus on returns rather than on prices?

- Scale-free
- Greater statistical stationarity over time
- Characteristic that portfolio managers want. Pay price to earn return.

Portfolio Math

Portfolios with holdings in N stocks:

- $q_{j,t}$ = # of **shares** of security j held in portfolio q at date t where $q_{j,t} > 0$ if long and $q_{j,t} < 0$ if short
- $q_{j,t} P_{j,t}$ = Dollar amount invested in portfolio q in stock j at date t
- $P_{q,t} = \sum_{j=1,\dots,N} q_{j,t} P_{j,t}$ = Total value of portfolio q at date t
- $w_{j,t} = \frac{q_{j,t} P_{j,t}}{\sum_{k=1,\dots,N} q_{k,t} P_{k,t}}$ = Portfolio **weight** = % of portfolio invested in security j at date t
 - Portfolio weights are “de-scaled” portfolio holdings.
 - Weights can change over time t in some strategies.
 - **Positive-investment** portfolios where $P_{q,t} > 0$: Weights sum to $\sum_{j=1,\dots,N} w_{j,t} = \mathbf{1}$
 - **Zero-investment** portfolios where $P_{q,t} = 0$ (long/short and arbitrage): Weights sum to **0** where **denominator** in zero-investment portfolio weights = total value of **long** positions (= short value).
- $CF_{q,t+1} = \sum_{j=1,\dots,N} q_{j,t} CF_{j,t+1}$ = Total dividends and other cash distributions received at $t+1$
- $P_{q,t+1} = \sum_{j=1,\dots,N} q_{j,t} P_{j,t+1}$ = Ex dividend portfolio resale value at date $t+1$

Portfolio Math (2)

Portfolio return is **weighted average** of individual asset returns:

$$r_{qt+1} = \frac{P_{qt+1} + CF_{qt+1} - P_{qt}}{P_{qt}} = \sum_{j=1, \dots, N} \frac{q_{jt} P_{jt+1} + q_{jt} CF_{jt+1} - q_{jt} P_{jt}}{P_{qt}} = \sum_{j=1, \dots, N} \frac{q_{jt} P_{jt}}{P_{qt}} \times \frac{P_{jt+1} + CF_{jt+1} - P_{jt}}{P_{jt}}$$

$$\rightarrow r_{qt+1} = \sum_{j=1, \dots, N} w_{jt} r_{jt+1}$$

Portfolio choice problem:

- Choose weights $w_{1t}, \dots, w_{Nt}$ to optimize an investor objective
- Individual holdings are “building blocks” used to construct portfolios
- One portfolio differs from another because their weights are different

Intuition:

- Choose weights to construct probability distribution from which portfolio returns will be drawn.

Examples:

- **Equally weighted:** $w_{jt} = 1/N$ (equal across holdings)
- **Constant mix:** $w_{jt} = w_{jt+1} = w_{jt+2} = \dots = w_j$ (constant over time but not equal across holdings)
- **Market-value (MV) weighted:** $w_{jt} = \frac{MV_{jt}}{\sum_{i=1, \dots, N} MV_{it}}$ where MV = price x amount outstanding

Illustration of Portfolio Return Calculations

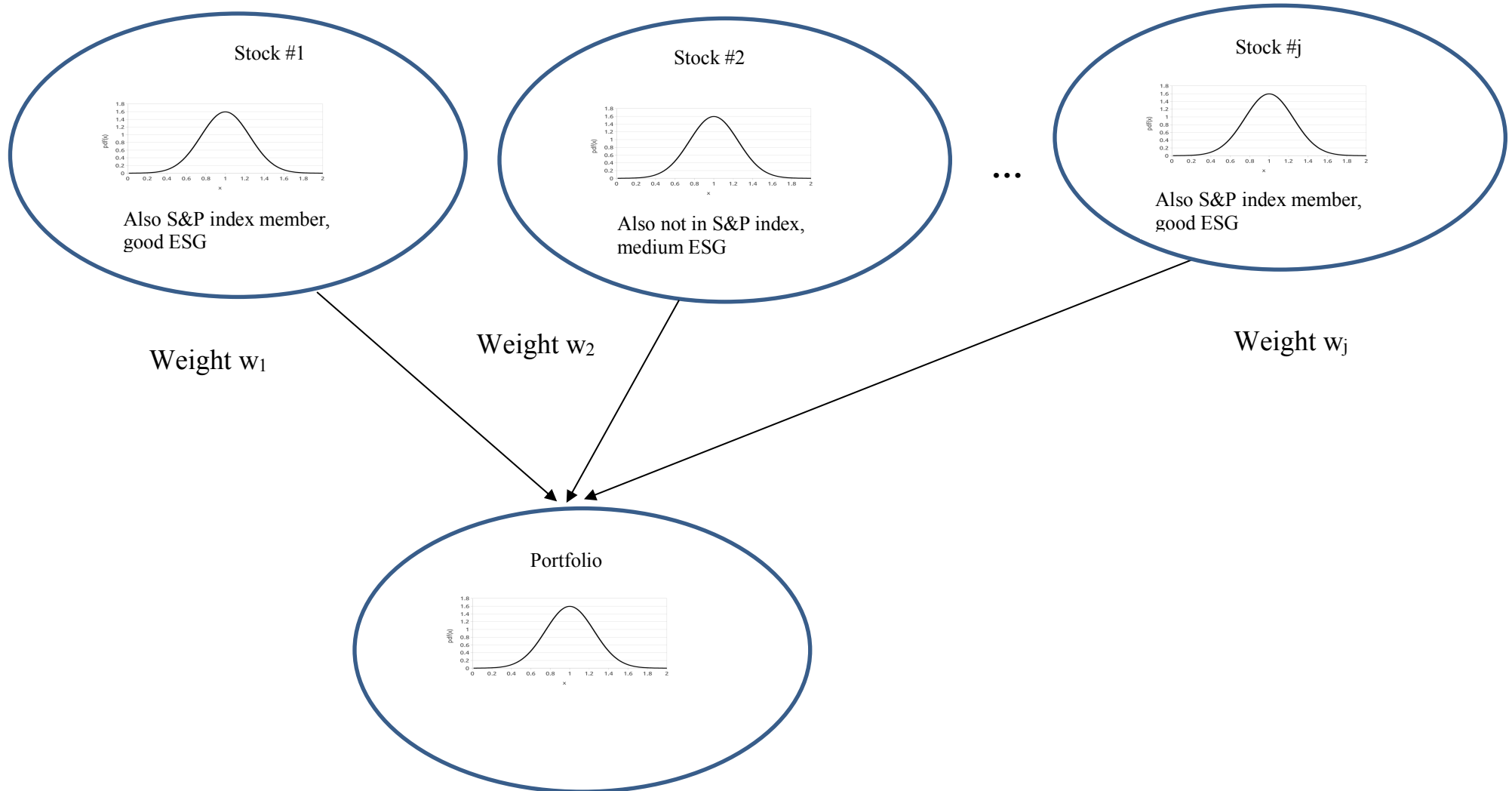
Illustration of Portfolio Return Calculations

Prepared by Duane Seppi (last revised 8/2018)

For educational use only.

Stocks	Current Prices	# Shares	Position Values	Portfolio Weights	Future Dividends	Future Prices	Stock Returns	Contribution to Port Return
1	\$21.50	500	\$10,750.00	0.346	\$0.50	\$22.00	0.047	0.016
2	\$75.17	300	\$22,551.00	0.726	\$0.00	\$75.10	-0.001	-0.001
3	\$50.00	-100	-\$5,000.00	-0.161	\$0.00	\$51.00	0.020	-0.003
4	\$2.75	1000	\$2,750.00	0.089	\$0.25	\$3.00	0.182	0.016
Totals:			\$31,051.00	1.000	Total:			0.028

Asset characteristics and portfolios characteristics



Portfolio Statistics

Ex post portfolio return is the random variable:

$$r_{q,t+1} = \sum_{j=1,\dots,N} w_{j,t} r_{j,t+1}$$

Expected return:

$$E[r_{q,t+1}] = \sum_{j=1,\dots,N} w_{j,t} E[r_{j,t+1}]$$

Variance:

$$\begin{aligned} \text{var}[r_{q,t+1}] &= \sum_{j=1,\dots,N} w_{j,t}^2 \text{var}[r_{j,t+1}] + \sum_{j=1,\dots,N} \sum_{i \neq j} w_{j,t} w_{i,t} \text{cov}(r_{j,t+1}, r_{i,t+1}) \\ &= \sum_{j=1,\dots,N} w_{j,t}^2 \text{var}[r_{j,t+1}] + \sum_{j=1,\dots,N} \sum_{i \neq j} w_{j,t} w_{i,t} \text{SD}(r_{j,t+1}) \text{SD}(r_{i,t+1}) \text{corr}(r_{j,t+1}, r_{i,t+1}) \end{aligned}$$

Conditional moments given information I_t at time t :

$$\begin{aligned} E[r_{q,t+1}] &\rightarrow E_{t-1}[r_{q,t+1}] \\ \text{var}[r_{q,t+1}] &\rightarrow \text{var}_{t-1}[r_{q,t+1}] \end{aligned}$$

Intuition:

- ❑ Statistics for individual securities are **exogenous**
- ❑ Construct **endogenous** portfolio statistics by **choice** of portfolio weights

Factor Structure of Portfolio Returns

Returns on a portfolio between $t-1$ and t

$$r_{p,t} = \sum_{j=1,\dots,N} w_{j,t-1} r_{j,t}$$

$$= \sum_{j=1,\dots,N} w_{j,t-1} [\alpha_j + \beta_{j,1} F_{1,t} + \beta_{j,2} F_{2,t} + \beta_{j,3} F_{3,t} + \dots + e_{j,t}]$$

$$= [\sum_{j=1,\dots,N} w_{j,t-1} \alpha_j] + [\sum_{j=1,\dots,N} w_{j,t-1} \beta_{j,1}] F_{1,t} + [\sum_{j=1,\dots,N} w_{j,t-1} \beta_{j,2}] F_{2,t} + \dots + [\sum_{j=1,\dots,N} w_{j,t-1} e_{j,t}]$$

$$= \alpha_{p,t-1} + \beta_{p,1,t-1} F_{1,t} + \beta_{p,2,t-1} F_{2,t} + \dots + e_{p,t}$$

- $\alpha_{p,t-1}$ = portfolio intercept = weighted average intercept $\sum_{j=1,\dots,N} w_{j,t-1} \alpha_{j,t-1}$
- $\beta_{p,k,t-1}$ = portfolio sensitivity to factor k = weighted average $\sum_{j=1,\dots,N} w_{j,t-1} \beta_{j,k}$
- $e_{p,t}$ = idiosyncratic exposure = weighted average exposure $\sum_{j=1,\dots,N} w_{j,t-1} e_{j,t}$

- **Question:** Why “t-1” subscripts on portfolio intercept & factor sensitivities but not stocks?

Portfolio manager's decision

- **Choose weights** $w_{1,t-1}, \dots, w_{N,t-1}$ for individual securities at time $t-1$ to achieve desired feasible portfolio factor characteristics (intercept, factor sensitivities, and idiosyncratic exposures) at time t .
- Don't need to know ex post realization of a factor F_{kt} to know ex ante sensitivity $\beta_{p,k,t-1}$ of a portfolio's return to that factor.

Examples of Factor-Based Portfolio Strategies

Managing with factor forecasts

- Don't want factor risk, then choose $w_1, \dots, w_N$ so factor exposure is zero: $\sum_{j=1, \dots, N} w_{j,t-1} \beta_{j,k} = 0$.
- If optimistic about factor, then want: $\sum_{j=1, \dots, N} w_{j,t-1} \beta_{j,k} > 0$.
- If pessimistic, then want: $\sum_{j=1, \dots, N} w_{j,t-1} \beta_{j,k} < 0$.

Benchmarking with factor forecasts

- Suppose pessimistic about factor $F_{k,t}$.
- If benchmarked to index with an exposure $\beta_{\text{index},k} > 0$, then unwilling to cut exposure to zero.
- Underweight portfolio exposure by, say, 25 percent: e.g., $\sum_{j=1, \dots, N} w_{j,t-1} \beta_{jk} = (1 - 0.25) \beta_{\text{index},k}$

Mimicking indices

- Portfolio is benchmarked relative to a given index (e.g., the S&P 500, EAFA, etc.).
- Identified stocks $1, \dots, N$ as potential “buy” candidates for an actively managed portfolio.
- Want to form portfolio which will “track” factor and yet benefit from your security selection.
- Statistical and mechanical approaches.

Investment Preferences

What statistical properties and other portfolio characteristics do investors like/dislike?

- ❑ More preferred to less? Like portfolios with higher expected returns $E[r_{qt+1}]$
- ❑ Risk aversion? Dislike portfolios with higher variances $\text{var}[r_{qt+1}]$
- ❑ Skewedness preferences? Other “higher” moments?
- ❑ Correlation with macro conditions (e.g., GDP, inflation)? With future investment opportunities?
- ❑ Correlation with personal risks (e.g., job loss, low wages, retirement, health, etc.)

Caution:

- ❑ Risk aversion need **not** imply investors dislike **individual** stocks with high variances. Why?

What else?

- ❑ Environmental, Social and Governance (ESG)
- ❑ Liquidity
- ❑ Legal and tax restrictions
- ❑ Sentimental value (e.g., company founders)

Investment management:

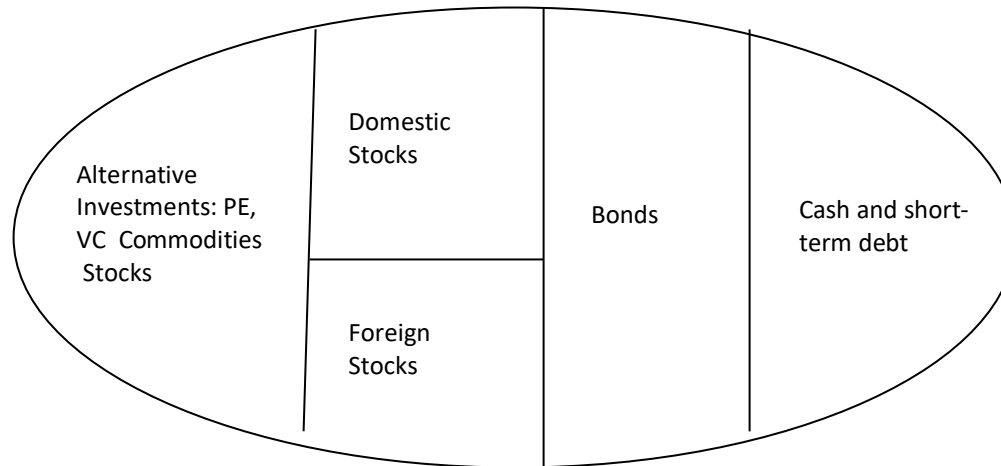
- ❑ Choose security holdings to design the optimal probability distribution for portfolio returns

Preferences may change over time

- ❑ With business cycle
- ❑ Sentiment (“Greed-Fear”) – Different from biased perceptions (over optimism or pessimism)

Asset Classes

Securities often classified in terms of asset classes



Investment choices within each asset class

- ❑ Hold securities directly or invest via managed portfolios (hedge fund, mutual fund, VC fund)
- ❑ Actively managed or passively indexed

Additional asset-class subdivisions:

- ❑ Large, midcap, and small capitalization stocks
- ❑ Growth versus value
- ❑ Industry
- ❑ Domestic and international bonds
- ❑ Treasury, corporate, asset-backed, tax-exempt bonds

Asset-Class Return Decompositions

Portfolio returns can be decomposed into:

- Asset class weights
- Returns on asset class sub-portfolios

$$r_{q,t+1} = \sum_{j=1,\dots,N} w_{j,t} r_{j,t+1}$$

$$= \sum_{\text{stocks}} w_{j,t} r_{j,t+1} + \sum_{\text{bonds}} w_{j,t} r_{j,t+1} + \sum_{\text{cash}} w_{j,t} r_{j,t+1}$$

$$= \left[\sum_{\text{stocks}} w_{j,t} \right] \sum_{j=\text{stock}} \frac{w_{jt}}{\sum_{k=\text{stock}} w_{kt}} r_{j,t+1} + \left[\sum_{\text{bonds}} w_{j,t} \right] \sum_{j=\text{bond}} \frac{w_{jt}}{\sum_{k=\text{bond}} w_{kt}} r_{j,t+1} \\ + \left[\sum_{\text{cash}} w_{j,t} \right] \sum_{j=\text{cash}} \frac{w_{jt}}{\sum_{k=\text{cash}} w_{kt}} r_{j,t+1}$$

$$= w_{\text{stocks},t} r_{\text{stock},t+1} + w_{\text{bonds},t} r_{\text{bonds},t+1} + w_{\text{cash},t} r_{\text{cash},t+1}$$

where

- $w_{\text{stocks},t}$, $w_{\text{bonds},t}$, $w_{\text{cash},t}$ = asset-class weights for stock, bond, and cash parts of portfolio
- $r_{\text{stock},t+1}$ = return on stock sub-portfolio (i.e., subportfolio weights $\frac{w_{jt}}{\sum_{k=\text{stock}} w_{kt}}$ sum to 1), etc.

Taxonomy of Investment Decisions

Strategic asset allocation

- Examples:
 - Ad hoc rules of thumb (60/40 splits)
 - **Mean-variance** for strategic asset-class allocation
 - To operationalize, use **estimated moments** based on **historical** data

Dynamic asset-class allocation

- Examples:
 - Trying to forecast asset-class returns and risks
 - **Return forecasting** using regression, time-series models, neural nets, ML & historical data
 - **Factor forecasting** using similar empirical tools
 - Changing asset-class weights given changes in investor risk-bearing willingness
 - **CPPI**

Security selection

- Examples:
 - Constant mix within asset classes
 - Indexation
 - Long-short portfolios based on anomalies. Buy (sell) estimated high (low) $E[\text{return}]$ assets.
 - **Mean-variance** analysis for security selection
 - But way **more** estimated parameters than for strategic asset allocation!

3 Investment Decisions

Brinson, Singer, and Beebower [1991] suggest investment returns reflect 3 investment decisions.

Strategic asset allocation:

- What are average asset-class weights over time?
 - Proxy for long-term “**strategic**” weights

Dynamic asset allocation:

- Do asset-class weights change over time?
 - **Constant passive:** Maintain constant strategic weights. But **trading** to maintain constant mix
 - **Changing passive:** Buy & hold weights (e.g., MV). No trading required but **change** over time
 - **Active:** Tactical market timing (trying to predict returns, risks), portfolio insurance (dynamic risk management)

Security selection:

- What is in asset-class sub-portfolios?
 - Passive: Indexation
 - Active: Security selection (i.e., stock picking)

Return attribution

- Assess contribution of active management along single dimensions, holding others fixed.

Interpretation

Formulas		Security Selection	
		Active	Indexed
Tactical AC Weights	Dynamic	$\sum_{i=1}^N w_i^D r_{i,t}^A$	$\sum_{i=1}^N w_i^D r_{i,t}^I$
	Constant	$\sum_{i=1}^N w_i^C r_{i,t}^A$	$\sum_{i=1}^N w_i^C r_{i,t}^I$

Example below		
	Active	Indexed
Dynamic	M: 0.0260 SD: 0.1340	M: 0.0465 SD: 0.1185
Constant	M: 0.0184 SD: 0.1348	M: 0.0470 SD: 0.1198

Questions

- ❑ Is the investor good at market-timing?
- ❑ Is the investor good at active security selection?
- ❑ What is the best investment strategy for this investor given what investor is good at?

Illustration of Return Attribution

Illustrated Brinson, Singer, and Beebower Decomposition

Prepared by Duane Seppi (last revised 8-2018)

For educational use only.

Monthly Returns (Not Annualized)							Index Returns	
Month	Stock A	Stock B	Stock C	Bond 1	Bond 2	t-bill	Stock	Bond
1	-0.1992	0.1518	-0.0513	-0.0164	0.0585	0.0013	0.0723	0.0100
2	-0.0344	-0.1925	-0.0199	-0.0059	0.0135	0.0013	-0.0542	0.0000
3	0.1092	0.1879	-0.0292	0.0177	-0.0791	0.0012	0.0668	0.0030
4	0.0146	-0.0908	-0.0591	0.0217	-0.0521	0.0011	0.0051	-0.0100
5	-0.0457	0.0317	-0.0460	-0.0209	-0.1193	0.0013	0.0250	-0.0300
6	0.1456	-0.0074	-0.0151	0.0069	0.0360	0.0013	-0.0107	0.0211
7	-0.0022	-0.1091	0.0074	0.0771	0.0370	0.0013	-0.0541	0.0400
8	-0.0478	-0.1275	-0.0947	-0.0169	-0.0010	0.0013	-0.0817	-0.0200
9	0.0189	0.0829	0.0745	0.0055	0.0284	0.0011	0.0181	0.0003
10	0.0903	0.0930	0.0341	0.0139	-0.0041	0.0013	0.0752	0.0300
11	0.0561	0.1375	0.0438	0.0112	0.0250	0.0013	0.0076	0.0050
12	0.0624	-0.0287	0.0438	-0.0268	-0.0429	0.0014	-0.0156	-0.0100
Annualized Return Statistics							Index Returns	
	Stock A	Stock B	Stock C	Bond 1	Bond 2	t-bill	Stock	Bond
Mean	0.1678	0.1287	-0.1116	0.0671	-0.1001	0.0156	0.0537	0.0394
SD	0.3165	0.4265	0.1756	0.0963	0.1867	0.0003	0.1780	0.0697

Return Attribution (2)

Client Security Holding Weights (at beginning of month)						
Month	Stock A	Stock B	Stock C	Bond 1	Bond 2	t-bill
1	0.20	0.10	0.30	0.15	0.15	0.10
2	0.21	0.10	0.30	0.13	0.15	0.11
3	0.25	0.11	0.30	0.19	0.18	-0.03
4	0.20	0.12	0.31	0.20	0.20	-0.03
5	0.20	0.10	0.32	0.13	0.15	0.10
6	0.25	0.13	0.31	0.12	0.15	0.04
7	0.20	0.10	0.31	0.11	0.15	0.13
8	0.23	0.10	0.33	0.07	0.15	0.12
9	0.23	0.10	0.32	0.07	0.13	0.15
10	0.22	0.13	0.33	0.10	0.15	0.07
11	0.20	0.14	0.33	0.10	0.15	0.08
12	0.23	0.14	0.33	0.12	0.15	0.03

Return Attribution (3)

Monthly Asset-Class Weights and Asset-Class Subportfolio Returns

Month	Stock Weight	Stock AC Return	Bond Weight	Bond AC Return	Cash Weight	Cash AC Return
1	0.60	-0.0667	0.30	0.0211	0.10	0.0013
2	0.61	-0.0532	0.28	0.0045	0.11	0.0013
3	0.66	0.0594	0.37	-0.0294	-0.03	0.0012
4	0.63	-0.0418	0.40	-0.0152	-0.03	0.0011
5	0.62	-0.0334	0.28	-0.0737	0.10	0.0013
6	0.69	0.0446	0.27	0.0231	0.04	0.0013
7	0.61	-0.0148	0.26	0.0540	0.13	0.0013
8	0.66	-0.0833	0.22	-0.0061	0.12	0.0013
9	0.65	0.0561	0.20	0.0204	0.15	0.0011
10	0.68	0.0635	0.25	0.0031	0.07	0.0013
11	0.67	0.0671	0.25	0.0195	0.08	0.0013
12	0.70	0.0354	0.27	-0.0357	0.03	0.0014

Strategic Weights

Stock	Bonds	t-bills
0.65	0.28	0.07

Annualized Asset-Class Subportfolio Return Statistics

	Stock	Bond	Cash	Index Returns	
				Stock	Bond
Mean	0.0329	-0.0145	0.0156	0.0537	0.0394
SD	0.1973	0.1169	0.0003	0.1780	0.0697

Return Attribution (4)

Brinson, Singer and Beebower Decompositions

Month	Client Port. Ret	Strategic Port . Ret.	Security Select only	Timing Only
1	-0.0336	0.0498	-0.0373	0.0465
2	-0.0310	-0.0350	-0.0331	-0.0329
3	0.0283	0.0442	0.0304	0.0452
4	-0.0324	0.0006	-0.0312	-0.0008
5	-0.0412	0.0080	-0.0421	0.0073
6	0.0371	-0.0010	0.0355	-0.0017
7	0.0052	-0.0238	0.0055	-0.0224
8	-0.0562	-0.0585	-0.0556	-0.0582
9	0.0407	0.0119	0.0421	0.0120
10	0.0441	0.0572	0.0422	0.0587
11	0.0499	0.0064	0.0490	0.0064
12	0.0152	-0.0128	0.0131	-0.0136

Annualized BSB Statistics

	Client Port. Ret	Strategic Port . Ret.	Security Select only	Timing Only
Mean	0.0260	0.0470	0.0184	0.0465
SD	0.1340	0.1198	0.1348	0.1185

Constant Proportion Portfolio Insurance (CPPI)

- Notation:

V_t = Total dollar **value** of a portfolio

E_t = **Dollar exposure** of active position

S_t = Dollar value of **safe** investment = $V_t - E_t$

F_t = Dollar value of a **minimum acceptable floor** (insurance level)

c_t = Dollar **cushion** of portfolio value over the floor = $V_t - F_t$

m = **Multiplier**

- **CPPI strategy:**

$E_t = mc_t$... sorry about that, Dr. Einstein!

- Active exposure = multiplier $\times$ cushion

- Payoff in “trending market”? If reversals (i.e., trading range market)

Generalizations and Examples:

Investor	Floor	Safe Asset	Active
Classic CPPI	Fixed dollar amount	t-bills	Stock
Pension with benefit cash flows in T (in 20 years)	PV of benefit in T	Long-dated bond maturing in T	Anything with an $E(\text{return}) > T\text{-}t$ bond $E(\text{return})$.
S&P 500 index fund	99 percent of random value if buy and hold S&P	S&P 500 index	Anything with $E(\text{return}) > \text{S\&P } E(\text{return})$

Do Funds Make More When They Trade More?

From Pastor, Stambaugh, and Taylor, *Journal of Finance* (2017), forthcoming

Sample:

- 3,126 actively managed U.S. domestic equity mutual funds over 1979–2011 period
- Monthly panel data (months t , funds i)

Model:

$$R_{i,t} = a_i + b_i X_{i,t-1} + \varepsilon_{i,t}$$

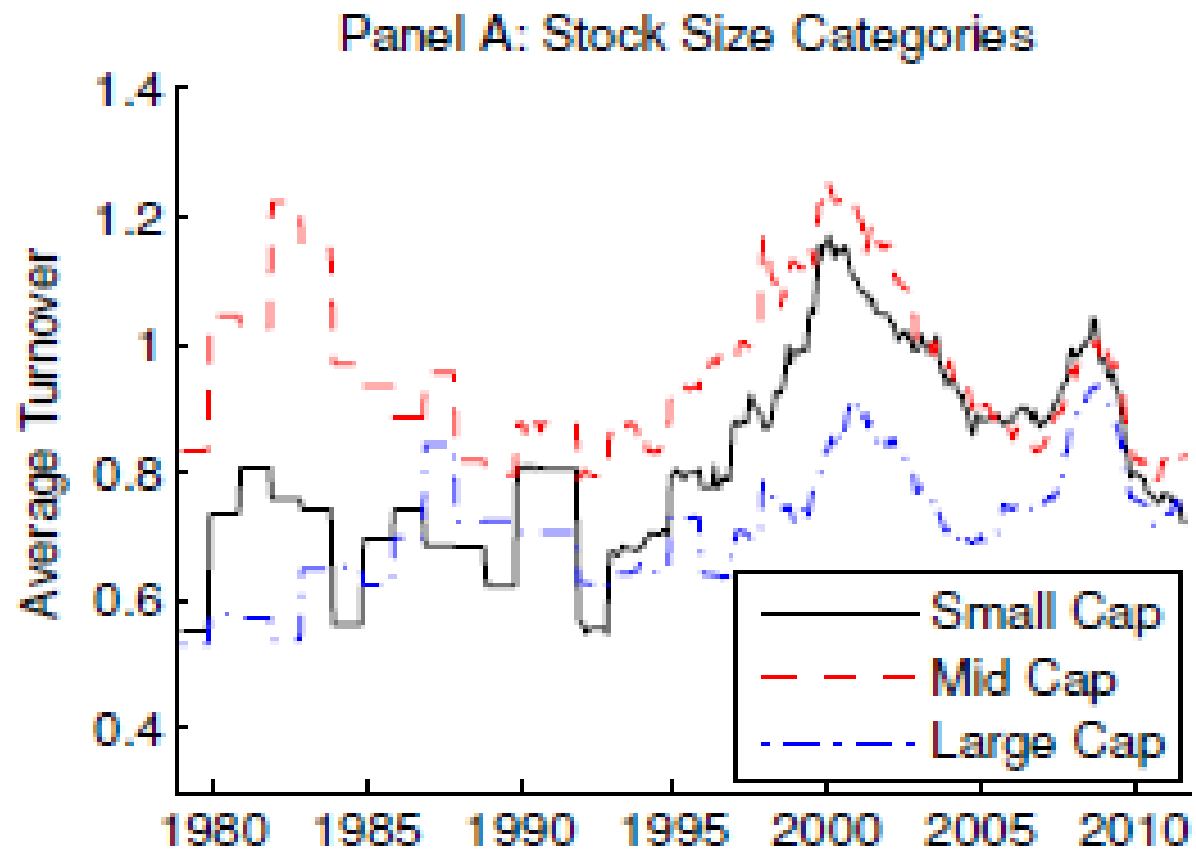
$R_{i,t}$ = gross excess return $r_{i,t} - r_{i,t}^b$ on fund i (before fees) in month t in excess of return on fund i 's MorningStar benchmark

$X_{i,t-1}$ = annual FundTurn $_{i,t-1} = \frac{\min \{TotalBuys_{i,t-1}, TotalSells_{i,t-1}\}}{average(TNA_{i,t-1})}$ from most recent SEC reporting year

Fixed effects: a_i can vary across firms

Restriction: $b_1 = b_2 = \dots = b$

Fund Turnover over Time



From: Pastor, Stambaugh, and Taylor (2017)

Regression Results

Table I
Turnover-Performance Relation in the Cross Section and Time Series

The table reports the estimated slope coefficients from four different panel regressions of $R_{i,t}$ on $FundTurn_{i,t-1}$. $R_{i,t}$ is fund i 's net return plus expense ratio minus Morningstar's designated benchmark return in month t . $FundTurn_{i,t-1}$ is fund i 's turnover for the most recent fiscal year that ends before month t . The four regressions differ only in their treatment of fixed effects. Heteroskedasticity-robust t -statistics clustered by sector $\times$ month are in parentheses, where "sector" is defined as Morningstar style category. Data are from 1979–2011. There are 282,738 fund-month observations in the panel.

Fund Fixed Effects	Month Fixed Effects	
	No	Yes
Yes	0.00125 (6.67)	0.00118 (7.08)
No	0.00043 (2.05)	0.00039 (2.04)

From: Pastor, Stambaugh, and Taylor (2017)

Take-Away

- Algebra of relation between portfolio returns & portfolio weights and individual asset returns
- Idea of preferences over portfolios
- Concept of asset classes
- 3 Investment decisions: Strategic, dynamic, and security selection
- Return attribution
- Trading and performance